Diskrete Mathematik

Woche 8 - HG G 26.5

https://github.com/tschai-yim/eth-resources

Feedback (anonym): https://forms.gle/xnkcjsdj86LK6yMh7



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Heutige Agenda

1. Teil (45 min)
- 20 min: Theorie-Übersicht mit Beispielen
- 25 min: Übungen

10 min Pause (hier persönliche Fragen)

2. Teil (45 min): Übungen

Theorie-Übersicht

Kapitel 4

Number Theory

- **Analyse properties of numbers** around divisibility and addition.
- We only look at small subset.

Divisibility

- **Divisor:** Integer a **divides** integer b ($a \mid b$) if $\boldsymbol{b = ac}$ for some integer c. [Def 4.1]
  - a: **divisor**, b: **multiple**.
- **Remainder:** For ints a, d with $d \neq 0$, unique integers q, r exist where: [Thm 4.1]
  - $a = q \cdot d + r$ and $0 \leq r < |d|$.
  - Notation: $R_d(a)$ or $a \bmod d$.
- **Greatest Common Divisor (GCD):** [Def 4.2]
  1. d is a common divisor ($d \mid a \wedge d \mid b$).
  2. All other common divisors divide d.
- **Least Common Multiple (LCM):**
  - Common multiple that divides all other common multiples. [Def 4.5]
  - Calculate via GCD: $\operatorname{lcm}(a, b) = \frac{|a \cdot b|}{\gcd(a,b)}$
- **Key Idea:** "**Greatest**" / "**Least**" refers to the **divisibility relation**, not $\leq$ order.

Euclidean Algorithm (optional?)

- **Principle:** $\gcd(a, b) = \gcd(b, a \bmod b)$. [Lem 4.2]
- **Procedure:** Repeatedly replace (a, b) with $(b, a \bmod b)$ until the remainder is 0. The last non-zero remainder is the GCD.
- **Example ($\gcd(252, 198)$):**
  1. $\gcd(252, 198) = \gcd(198, 252 \bmod 198) = \gcd(198, 54)$
  2. $\gcd(198, 54) = \gcd(54, 198 \bmod 54) = \gcd(54, 36)$
  3. $\gcd(54, 36) = \gcd(36, 18)$
  4. $\gcd(36, 18) = \gcd(18, 0) \Rightarrow \gcd = 18$.

Ideals

- **Definition:** Ideal (a, b): set of all integer linear combinations of a, b. [Def 4.4]
  - $(a, b) = \{ua + vb \mid u, v \in \mathbb{Z}\}$.
- $\boldsymbol{(a, b) = (\gcd(a, b))}$ (Ideal (a, b) is all multiples of $\gcd(a, b)$) [Lem 4.3, Lem 4.4]
- **Bézout's Identity:** u, v exist s.t. $\gcd(a, b) = ua + vb$. [Cor 4.5]
  - **Extended Euclidean Algorithm (optional?):** Work backwards from the standard algorithm's steps.
    - **Example:** Find u, v for $\gcd(252, 198) = 18$:
      1. Equations from Euclidean algorithm:
         $54 = 252 - 1 \cdot 198$; $36 = 198 - 3 \cdot 54$; $18 = 54 - 1 \cdot 36$.
      2. Substitute backwards: $18 = 54 - 1 \cdot (198 - 3 \cdot 54) = 4 \cdot 54 - 1 \cdot 198$.
      3. Continue: $18 = 4 \cdot (252 - 1 \cdot 198) - 1 \cdot 198 = 4 \cdot 252 - 5 \cdot 198$.
      4. Result: $u = 4, v = -5$.

Prime Numbers & Factorization

- **Prime Numbers:** Positive int $p > 1$ with only positive divisors 1 and p. [Def 4.6]
  - **Infinitude:** There are infinitely many primes. [Thm 4.9]
  - **Gaps:** Can be arbitrarily large. [Thm 4.10]
  - **Density:** The prime-counting function $\pi(x)$ counts primes $\leq x$. [Def 4.7]
- **Composite Numbers:** An integer > 1 that is not prime. [Def 4.6]
  - **Testing:** Trial division by primes up to $\sqrt{n}$. [Lem 4.12]
- **Fundamental Theorem of Arithmetic:** Every integer $n > 1$ has a **unique** prime factorization. [Thm 4.6]
  - For $a = \prod p_i^{e_i}$ and $b = \prod p_i^{f_i}$:
    - $\gcd(a, b) = \prod p_i^{\min(e_i, f_i)}$
    - $\operatorname{lcm}(a, b) = \prod p_i^{\max(e_i, f_i)}$
  - **Identity:** $\gcd(a, b) \cdot \operatorname{lcm}(a, b) = |a \cdot b|$ because $\min(e, f) + \max(e, f) = e + f$.

Modular Arithmetic

- **Congruence Relation:** $a \equiv b \bmod m$ if $m \mid (a - b)$. [Def 4.8]
  - An **equivalence relation**. [Lem 4.13]
  - Equivalent to $R_m(a) = R_m(b)$. [Lem 4.16]
- **Properties:**
  - Compatible with addition and multiplication. [Lem 4.14]
  - For any polynomial f, $f(a_1, ...) \underset{m}{\equiv} f(R_m(a_1), ...)$. [Cor 4.15]
- **The Ring $\mathbb{Z}_m$:**
  - Set of remainders $\{0, ..., m-1\}$ with modular addition/multiplication.
  - If an equation has no solution in $\mathbb{Z}_m$, it has no integer solution in $\mathbb{Z}$.

Solving Systems of Congruences

- **Multiplicative Inverse:** Integer x where $ax \underset{m}{\equiv} 1$. [Def 4.9]
  - Exists iff $\gcd(a, m) = 1$. [Lem 4.18]
  - Found via Extended Euclidean Algorithm.
- **Chinese Remainder Theorem (CRT):**
  - Solves a system $x \equiv a_i \bmod m_i$ for pairwise coprime moduli m_i. [Thm 4.19]
  - Has **unique solution** modulo $M = \prod m_i$.
  - **Solution Formula:** $x \equiv \sum_{i=1}^{r} a_i M_i N_i \bmod M$, where:
    1. $M = \prod m_i$
    2. $M_i = \frac{M}{m_i}$
    3. N_i = inverse of $M_i \bmod m_i$.

Calculating CRT

From dcamenisch's Cheatsheet:

Ex. $x \equiv_3 2$, $x \equiv_4 1$, $x \equiv_5 4$, $M = 60$

1. $M_i = \frac{M}{m}$: $M_1 = 20$, $M_2 = 15$, $M_3 = 12$

2. $N_i M_i \equiv_{m_i} 1$: $N_1 \cdot 20 \equiv_3 1 \Rightarrow N_1 = 2$, $N_2 \cdot 15 \equiv_4 1 \Rightarrow N_2 = 3$, $N_3 \cdot 12 \equiv_5 1 \Rightarrow N_3 = 3$

3. $\sum_{i=1}^{r} a_i \cdot M_i \cdot N_i = 2 \cdot 20 \cdot 2 + 1 \cdot 15 \cdot 3 + 4 \cdot 12 \cdot 3 \equiv_M \underline{\underline{29}}$

If we have something like $x \equiv_{12} 8$, $x \equiv_{15} 2$ we need to decompose since $\gcd(12, 15) \neq 1$.

We get: $x \equiv_3 2$, $x \equiv_4 0$ and $x \equiv_3 2$, $x \equiv_5 2$

need to be equal, else no solution

$2x^2 + 8 \equiv_{13} 6 \Rightarrow 2x^2 \equiv_{13} -2 \equiv_{13} 11 \equiv_{13} 24 \Rightarrow x^2 \equiv_{13} 12 \equiv_{13} 25 \Rightarrow x \equiv_{13} \pm 5$

Applications in Cryptography

- **One-Way Function:** Easy to compute, hard to reverse.
- **Discrete Logarithm Problem (DLP):** Given y, g, p, find x in $y \equiv g^x \bmod p$. Believed hard.
- **Diffie-Hellman Key Exchange:** Establishes shared secret over public channel.
  1. **Public:** Prime p, base g.
  2. **Alice:** Secret $a \to$ public $A = g^a \bmod p$.
  3. **Bob:** Secret $b \to$ public $B = g^b \bmod p$.
  4. **Shared Secret:** $S = B^a \equiv A^b \equiv g^{ab} \bmod p$.

Übungen

https://github.com/tschai-yim/eth-resources unter
`03_TA/26HS_252-0025-01L_Diskrete_Mathematik/exercises`

Problem 1: Divisibility Relations

Prove or disprove for all $a, b, c \in \mathbb{Z}$.

1. $a \mid b \wedge b \mid a \Rightarrow a = b$
2. $a \mid bc \Rightarrow a \mid b \vee a \mid c$
3. $a \mid b \Rightarrow |a| \leq |b|$

Def 4.1 For integers a and b, a **divides** b ($a \mid b$) if there exists an integer c such that:

$$b = ac$$

Def 4.6 A positive integer $p > 1$ is called **prime** if the only positive divisors of p are 1 and p.

Lem 4.7 If a prime p divides a product $x_1 x_2 \cdots x_n$, then:

$$p \mid x_i \quad \text{for some} \quad i \in \{1, \ldots, n\}$$

Problem 2: Division with Remainder

1. Determine q and r from Theorem 4.1 for $(a, d) \in \{(-17, 5), (17, -5), (-17, -5)\}$.
2. Prove or disprove for all $a \in \mathbb{Z}$ and $d \geq 1$: $R_d(-a) = d - R_d(a)$.

Thr 4.1 For all integers a and $d \neq 0$, there exist unique integers q and r satisfying:

$$a = dq + r \quad \text{and} \quad 0 \leq r < |d|$$

The remainder r is denoted by $R_d(a)$.

Def 4.1 For integers a and b, a divides b ($a \mid b$) if there exists an integer c such that:

$$b = ac$$

Problem 3: Linear Combinations and $\gcd$

Let $a, b, k \in \mathbb{Z}$ with a, b not both 0, and $n \in \mathbb{N}$.

1. Prove: $\gcd(a + kb, b) = \gcd(a, b)$.
2. Determine $\gcd(n^2 + 1, n + 1)$.

Def 4.1 For integers a, b, $a \mid b$ if there exists $c \in \mathbb{Z}$ such that $b = ac$.

Def 4.2 / 4.3 For $a, b \in \mathbb{Z}$ (not both 0), $\gcd(a, b)$ is the unique positive integer d satisfying:

1. $d \mid a \wedge d \mid b$
2. $\forall c \in \mathbb{Z} : (c \mid a \wedge c \mid b) \Rightarrow c \mid d$

Lem 4.2 For any integers m, n, and q:

$$\gcd(m, n - qm) = \gcd(m, n)$$

Problem 4: Bézout's Identity and EEA

Find $u, v \in \mathbb{Z}$ with $56u + 15v = 1$.

Thr 4.1 For all integers a and $d \neq 0$, there exist unique integers q, r such that:

$$a = dq + r \quad \text{and} \quad 0 \leq r < |d|$$

Def 4.4 The ideal generated by $a, b \in \mathbb{Z}$ is the set of all integer linear combinations:

$$(a, b) = \{ua + vb \mid u, v \in \mathbb{Z}\}$$

Lem 4.2 For any integers m, n, and q:

$$\gcd(m, n - qm) = \gcd(m, n)$$

Cor 4.5 For integers a, b (not both 0), there exist $u, v \in \mathbb{Z}$ such that:

$$\gcd(a, b) = ua + vb$$

Problem 5: Divisibility and Coprimality

Prove or disprove for all $a, b, c \in \mathbb{Z}$: $a \mid c \land b \mid c \Rightarrow ab \mid c$. If the claim is false, repair it with an additional hypothesis.

Def 4.3 Two integers a and b (not both 0) are called **relatively prime** (or coprime) if:

$$\gcd(a, b) = 1$$

Def 4.5 The **least common multiple** $l = \operatorname{lcm}(a, b)$ is the common multiple dividing every other common multiple:

$$a \mid l \land b \mid l \land \forall m((a \mid m \land b \mid m) \Rightarrow l \mid m)$$

Cor 4.5 For $a, b \in \mathbb{Z}$ (not both 0), there exist $u, v \in \mathbb{Z}$ such that:

$$\gcd(a, b) = ua + vb$$

Ref For all positive integers a, b:

$$\gcd(a, b) \cdot \operatorname{lcm}(a, b) = a \cdot b$$

Problem 6: Mersenne Primes

Prove: if $2^n - 1$ is prime ($n \geq 1$), then n is prime. Is the converse true?

Def 4.6 A positive integer $p > 1$ is called **prime** if the only positive divisors of p are 1 and p. An integer greater than 1 that is not prime is called **composite**.

Thr 4.6 Every positive integer can be written uniquely (up to the order of factors) as the product of primes.

Lem 4.12 Every composite integer n has a prime divisor $p \leq \sqrt{n}$.

Problem 7: Infinitude of Primes Modulo 4

Prove: there are infinitely many primes p with $R_4(p) = 3$. Why does the same argument fail for $R_4(p) = 1$?

Thr 4.9 There are infinitely many primes.

Thr 4.1 For all integers a and $d \neq 0$, there exist unique integers q, r such that:

$$a = dq + r \quad \text{and} \quad 0 \leq r < |d|$$

Lem 4.7 If a prime p divides a product $x_1 x_2 \cdots x_n$, then:

$$p \mid x_i \quad \text{for some} \quad i \in \{1, \ldots, n\}$$

Lem 4.14 If $a \underset{m}{\equiv} b$ and $c \underset{m}{\equiv} d$, then:

$$a + c \underset{m}{\equiv} b + d \quad \text{and} \quad ac \underset{m}{\equiv} bd$$

Problem 8: Cancellation in Modular Congruences

Prove or disprove for all $a, b, c \in \mathbb{Z}$ and $m \geq 1$: $ac \underset{m}{\equiv} bc \wedge c \not\underset{m}{\equiv} 0 \Rightarrow a \underset{m}{\equiv} b$. If the claim is false, repair it with an additional hypothesis on c.

Def 4.8 For $a, b, m \in \mathbb{Z}$ with $m \geq 1$, a is **congruent** to b modulo m ($a \underset{m}{\equiv} b$) if:

$$m \mid (a - b)$$

Def 4.9 If $\gcd(a, m) = 1$, the unique solution $x \in \mathbb{Z}_m$ to the congruence $ax \underset{m}{\equiv} 1$ is the **multiplicative inverse** of a modulo m, denoted a^{-1} or $\frac{1}{a}$.

Lem 4.14 If $a \underset{m}{\equiv} b$ and $c \underset{m}{\equiv} d$, then:

$$ac \underset{m}{\equiv} bd$$

Lem 4.18 The congruence equation $ax \underset{m}{\equiv} 1$ has a solution $x \in \mathbb{Z}_m$ if and only if $\gcd(a, m) = 1$. The solution is unique.

Problem 9: Powers and Modular Reduction

Compute $R_{100}(3^{2026})$.

Lem 4.16 For any $a, b, m \in \mathbb{Z}$ with $m \geq 1$:

$$a \underset{m}{\equiv} R_m(a) \quad \text{and} \quad a \underset{m}{\equiv} b \Leftrightarrow R_m(a) = R_m(b)$$

Lem 4.14 If $a \underset{m}{\equiv} b$ and $c \underset{m}{\equiv} d$, then $ac \underset{m}{\equiv} bd$. In particular:

$$a^k \underset{m}{\equiv} b^k \quad \text{for all} \quad k \in \mathbb{N}$$

Cor 4.15 Let $f(x_1, ..., x_k)$ be an integer polynomial and $m \geq 1$. If $a_i \underset{m}{\equiv} b_i$ for all i:

$$f(a_1, ..., a_k) \underset{m}{\equiv} f(b_1, ..., b_k)$$

Cor 4.17 For any integer polynomial $f(x_1, ..., x_k)$ and $m \geq 1$:

$$R_m(f(a_1, ..., a_k)) = R_m(f(R_m(a_1), ..., R_m(a_k)))$$

Problem 10: Solvability of Diophantine Equations

Prove that $x^3 + y^3 + z^3 = 2030$ has no integer solutions.

Ref If an integer polynomial equation $f(x_1, ..., x_k) = 0$ has a solution in $\mathbb{Z}$, then it has a solution modulo m for every integer $m \geq 1$.

Cor 4.15 Let $f(x_1, ..., x_k)$ be an integer polynomial and $m \geq 1$. If $a_i \underset{m}{\equiv} b_i$ for all i:

$$f(a_1, ..., a_k) \underset{m}{\equiv} f(b_1, ..., b_k)$$

Cor 4.17 For any integer polynomial $f(x_1, ..., x_k)$ and $m \geq 1$:

$$R_m(f(a_1, ..., a_k)) = R_m(f(R_m(a_1), ..., R_m(a_k)))$$

Lem 4.16 For any $a, b, m \in \mathbb{Z}$ with $m \geq 1$:

$$a \underset{m}{\equiv} b \Leftrightarrow R_m(a) = R_m(b)$$

Problem 11: Wilson's Theorem

Prove: $(p-1)! \underset{p}{\equiv} -1$ for every prime p.

Def 4.6 A positive integer $p > 1$ is called **prime** if the only positive divisors of p are 1 and p.

Def 4.9 If $\gcd(a, m) = 1$, the unique solution $x \in \mathbb{Z}_m$ to $ax \underset{m}{\equiv} 1$ is the **multiplicative inverse** of a modulo m.

Lem 4.18 The congruence equation $ax \underset{m}{\equiv} 1$ has a solution $x \in \mathbb{Z}_m$ if and only if $\gcd(a, m) = 1$. The solution is unique.

Lem 4.7 If a prime p divides a product $x_1 x_2 \cdots x_n$, then:

$$p \mid x_i \quad \text{for some} \quad i \in \{1, \ldots, n\}$$

Problem 12: Systems of Modular Congruences

Find all solutions of $x \underset{6}{\equiv} 5$, $x \underset{10}{\equiv} 3$, $x \underset{15}{\equiv} 8$ with $0 \leq x < 30$, and justify that there are no others.

For which $c \in \mathbb{Z}_{15}$ does the system with $x \underset{15}{\equiv} c$ instead of $x \underset{15}{\equiv} 8$ have a solution?

Thr 4.19 Let $m_1, \ldots, m_r$ be pairwise relatively prime integers and $M = \prod_{i=1}^{r} m_i$. For every list $a_1, \ldots, a_r$ with $0 \leq a_i < m_i$, the system

$$x \underset{m_i}{\equiv} a_i \quad (1 \leq i \leq r)$$

has a unique solution x satisfying $0 \leq x < M$.

Ref For integers $m, n \geq 1$ with $\gcd(m, n) = 1$:

$$x \underset{mn}{\equiv} a \Leftrightarrow \left(x \underset{m}{\equiv} a \quad \text{and} \quad x \underset{n}{\equiv} a \right)$$

Def 4.8 For $a, b, m \in \mathbb{Z}$ with $m \geq 1$:

$$a \underset{m}{\equiv} b \Leftrightarrow m \mid (a - b)$$

Problem 13: Diffie-Hellman Key Exchange

Alice and Bob run the Diffie-Hellman protocol (Figure 4.1) with $p = 23$ and $g = 5$.

1. Alice chooses $x_A = 6$ and Bob chooses $x_B = 15$. Compute y_A, y_B, k_{AB}, k_{BA}.
2. Eve sees only y_A and y_B. Compute k_{AB} the way Eve would for such a small p.

Ref Public parameters: prime p, base g.

- Private keys: $x_A, x_B \in \{1, \ldots, p-1\}$
- Public keys: $y_A = R_p(g^{x_A})$, $y_B = R_p(g^{x_B})$
- Shared key:

$$k_{AB} = R_p\left(y_B^{x_A}\right) = R_p\left(y_A^{x_B}\right) = R_p(g^{x_A x_B})$$

Ref Given a prime p, base g, and value $y \in \mathbb{Z}_p$, find an integer x such that:

$$g^x \underset{p}{\equiv} y$$

Cor 4.17 For any polynomial (and power) with integer coefficients:

$$R_m\left(a^k\right) = R_m\left(\left(R_m(a)\right)^k\right)$$

Offene Fragen & Feedback

- Gibt es noch Fragen zu den heutigen Themen oder den Übungen?
- Habt ihr Feedback zur heutigen Übungsstunde?
  - Anonymes Formular: https://forms.gle/xnkcjsdj86LK6yMh7
- Bei Fragen: Schreibt mir eine E-Mail oder auf WhatsApp.

Schöne Woche und bis nächsten Montag!